

Constant rate of cutting of flux $\Rightarrow$ constant emf induced between centre and rim.

25 Ans: D

By Lenz's law, since the external B-field increases, the induced B setup by the induced current will be in opposite direction to the external field. At the front of the loop, current will be going upwards $\Rightarrow$ use this to deduce direction of induced current.

Use $\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{N(\Delta B)(A)}{\Delta t}$ to get the magnitude of the induced emf.

26 Ans: B